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Assessing ChatGPT adoption in Jordanian medical education: a UTAUT model approach

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Abstract

Background ChatGPT has shown significant promise in transforming medical education by streamlining research and improving teaching methods. However, its adoption in Middle Eastern medical education has remained underexplored. This study investigated the factors influencing the adoption of ChatGPT in Jordanian medical education using a modified Unified Theory of Acceptance and Use of Technology (UTAUT) framework.

Methods A cross-sectional survey was conducted with medical students and faculty members at the University of Jordan. A validated 33-item questionnaire distributed online and on campus targeted individuals familiar with the ChatGPT. Structural equation modeling (SEM) assessed the relationships between key constructs, including Performance Expectancy (PE), Effort Expectancy (EE), Perceived Risk (PR), Facilitating Conditions (FC), and attitude (ATT).

Results Among 127 participants (53% male, mean age 23.2 ± 7.6), ATT was significantly influenced by PE and EE, explaining 37% of its variance. Behavioral Intention (BI) was predicted by ATT and had a significant positive effect on actual usage. FC did not significantly influence EE or BI, suggesting a limited reliance on external support. Contrary to expectations, PR did not negatively affect ATT, indicating that utility outweighed concerns about misinformation or privacy. Overall, the model explained 53% of the variance in BI and 36.5% of the variance in actual usage.

Conclusion The adoption of ChatGPT in Jordanian medical education is driven by perceived utility and ease of use, with attitudes playing a pivotal role. Addressing misinformation risks and improving trust through tailored strategies can foster broader integration of AI tools, such as ChatGPT, in medical training.

Keywords Artificial intelligence, ChatGPT, Jordan, Medical education, Medical students

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Introduction

The emergence of ChatGPT has significantly advanced conversational AI. This adaptable AI system can generate content and code, summarizing information, engaging in creative writing, and executing style transformations. With training on vast multilingual datasets, it produces responses that mimic human conversation. The medical field has shown interest in AI applications, exploring their potential in clinical decision making, diagnosis, patient management, electronic health records, and medical education [1, 2].

The adoption of AI in medical education has enhanced learning methodologies, improved assessment accuracy, and supported clinical decisions. For instance, ChatGPT-4 demonstrated impressive accuracy in the United States Medical Licensing Examination (USMLE), with scores of 85.7% in Step 1, 92.5% in Step 2, and 86.9% in Step 3 [3]. This finding suggests its potential as a valuable tool for medical students.

AI-driven tools also support medical imaging training by refining anatomical learning through advanced image segmentation, thereby significantly contributing to educational outcomes [4, 5]. Natural Language Processing (NLP), including ChatGPT, has shown promise in delivering human-like language responses, but challenges in accuracy, interpretability, and ethical concerns persist [6–8].

ChatGPT's capabilities extend to streamlining literature reviews, analyzing extensive medical datasets, overcoming language barriers, and facilitating personalized, problem-based education [9, 10]. Additionally, ChatGPT shows promise in developing new drugs, improving patient-centered care in radiology and other domains, and serving as a reference for healthcare providers, such as in determining imaging steps in breast cancer screening [11–14].

Despite this promise, substantial limitations exist, notably the generation of inaccurate responses, biases, and potential misinformation, which can compromise the educational effectiveness and clinical decisions [7, 8, 15]. Furthermore, privacy concerns and trust issues present critical barriers that necessitate careful management to ensure ethical use of AI tools [16, 17].

In Middle Eastern contexts, particularly Jordan, AI adoption faces unique challenges, such as inadequate Arabic language optimization, faculty resistance, and limited integration into existing curricula, highlighting significant regional barriers [7, 18, 19]. Despite these critical challenges, there is limited research specifically addressing the adoption of AI, including ChatGPT, in Jordanian medical education, representing a clear knowledge gap.

To address this gap, this study employed a modified UTAUT framework to investigate the barriers and facilitators influencing ChatGPT adoption in Jordanian medical education, aiming to provide actionable insights to enhance its effective integration into medical curricula.

Materials and methods

Development of hypothesis and conceptual design

The UTAUT model was identified as the most suitable framework for assessing the crucial factors influencing users' willingness to adopt ChatGPT in medical education. This model, originally developed by Venkatesh et al., integrates eight prominent theoretical frameworks of technology acceptance, making it one of the most comprehensive and widely validated models in the field [20, 21]. The UTAUT model was particularly implemented in this study due to its ability to capture the multifaceted nature of ChatGPT adaptation in educational settings, providing a framework for understanding factors the influence the adoption of tools and ensures a comprehensive analyzation of both driver and barriers to the adaptation of ChatGPT, making it highly relevant for our study [22]. Chatterjee and Bhattacharjee propose the adaptation of the UTAUT model used in this study.

The UTAUT model excels in its ability to account for user behavior towards technology by incorporating key constructs Table 1 and is depicted in Fig. 1. It has been shown to explain up to 70% of the variance in Behavioral Intention and technology use, outperforming earlier models in predictive accuracy [20, 21]. For this study, Social Influence was excluded due to its limited relevance to medical students' adoption of ChatGPT, as supported

Table 1 Constructs and hypotheses

Construct	Definition	Hypothesis	References
Perceived Risk (PR)	Users' concerns about security, privacy, and credibility of information.	H1: PR negatively impacts ATT.	20, 22
Performance Expectancy (PE)	The degree to which individuals believe ChatGPT will enhance their educational performance.	H2: PE positively impacts ATT.	19, 23–25
Effort Expectancy (EE)	The perceived ease of using ChatGPT.	H3: EE positively impacts ATT.	19, 25–26
Facilitating Conditions (FC)	The availability of organizational and technical support for ChatGPT adoption.	H4a: FC positively impacts BI. H4b: FC positively impacts EE.	19, 27
Attitude (ATT)	Positive or negative beliefs about ChatGPT adoption, influencing Behavioral Intention.	H5: ATT positively impacts BI.	21, 27–28
Behavioral Intention (BI)	The intention to adopt ChatGPT, predicting actual use.	H6: BI positively impacts ChatGPT adoption.	25, 27–29

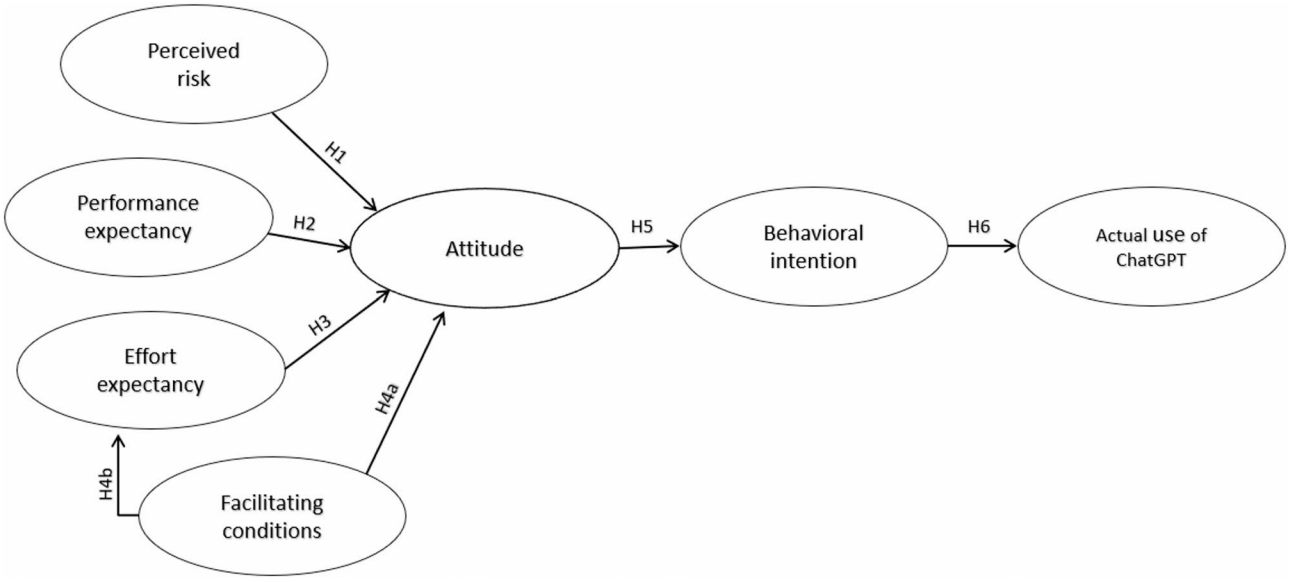

Fig. 1 Hypotheses of integrated UTAUT model

by findings from related studies. Instead, the focus was directed towards the remaining three constructs. Additionally, ATT was included as a mediating factor, based on its critical role in shaping Behavioral Intention in previous research [23, 24].

Given the specific context of medical education, the quality of knowledge provided by ChatGPT was identified as a key concern. Therefore, Perceived Risk (PR) was introduced as an exogenous factor affecting user adoption. This aligns with prior studies that highlight the negative influence of perceived risks, such as security, privacy, and credibility concerns, on technology adoption [23].

To streamline the analysis and focus on direct effects, moderators such as age, gender, experience, and voluntariness were excluded, based on the assumption that these factors have minimal impact on medical students' attitudes toward adopting ChatGPT [23]. The resulting modified UTAUT model posits that PR, PE, and EE influence BI through ATT. Facilitating Conditions were hypothesized to have both a direct effect on BI and an indirect effect on EE.

Study sample

This cross-sectional study utilized a validated 33-item questionnaire, originally developed and published in a previous study [25], which was distributed via email and QR codes on campus to minimize sampling bias. The study targeted medical students and professors at the University of Jordan and its affiliated hospital. Undergraduate medical students across all years (first to sixth) and professors of varying specialties participated (Supplementary File 1 shows the questionnaire).

We received Institutional Review Board (IRB) approval from the University of Jordan Medical College. Data were collected from April 8 to June 3, yielding 441 responses. To focus on participants familiar with ChatGPT, we excluded 300 participants who hadn't used ChatGPT and 14 participants who did not consent.

Statistical analysis

SEM was performed using SPSS AMOS version 26 to assess relationships among latent variables and validate the model's structure. SEM was selected for its ability to verify theoretical models and test relationships between constructs. The model's explanatory power was evaluated using R^2 values, indicating the variance in the dependent variable explained by independent variables [20]. FC were measured to assess the extent of the adoption of ChatGPT among organizations. Path coefficient in the SEM model illustrated the strength and direction of the r4elashionship between constructs. The R^2 values indicate the proportion of variance in the dependent variable explained by independent variables, to provide insight into the model's explanatory power.

Demographic data were analyzed using SPSS version 26. Continuous variables (e.g., age) were summarized using mean (± standard deviation), while categorical variables (e.g., gender) were reported as frequencies and percentages.

Results

A total of 442 individuals consented to participate in the study, with 127 respondents (28.73%) had prior experience of using the ChatGPT and were subsequently included in the analysis. The cohort comprised medical

Table 2 Characteristics of the study respondents

Demographic data		
Variable		N (%)
Gender		
Female		60(47)
Male		67(53)
Age		
18–20 years		35(27.6)
21 23 years		76(59.8)
> 23 years		16(12.6)
Teaching staff		11(8.7)
Medical students' year of study	1st	13(10.2)
	2nd	11(8.7)
	3rd	29(22.8)
	4th	47(37)
	5th	9(7.1)
	6th	7(5.5)

students and teaching staff (Table 2). The participants' ages ranged from 18 to 58 years, with an average of 23.2 years (SD = 7.6). The gender distribution was nearly balanced, with 67 males (53%) and 60 females (47%).

Structural equation modeling yielded seven hypotheses concerning the technology acceptance model, as depicted in Fig. 2 and summarized in Table 3. Key findings include that PE and EE both had significant positive effects on ATT (β = 0.344, p <.001; β = 0.428, p <.001, respectively), accounting for 37% of the variance in ATT. FC did not have a significant impact on EE (β = − 0.032, ns) or on BI (β = − 0.098, ns). ATT significantly influenced BI, demonstrating a strong positive relationship (β = 0.84, p <.001). The path from BI to Actual Use of ChatGPT (ACME) was also significant (β = 0.575, p <.001), with BI explaining 53% of the variance in ACME.

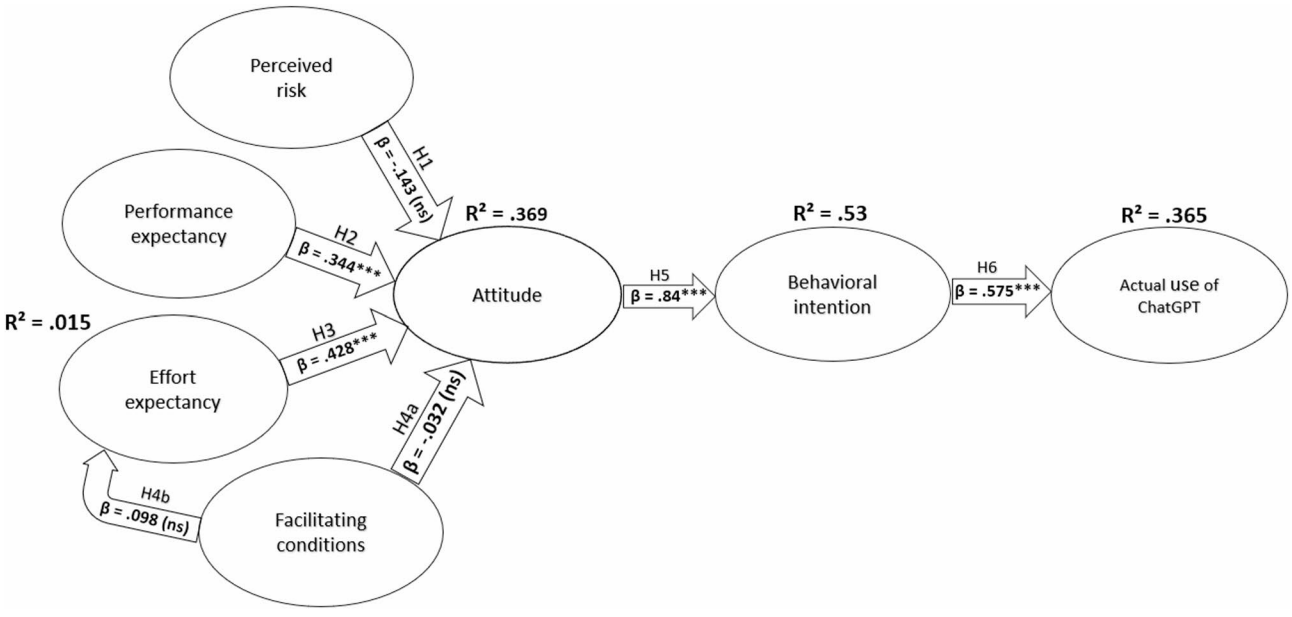

Fig. 2 Schematic illustration of the structural equation modeling (SEM). This figure shows the relationships between model constructs along with path coefficients and levels of significance. Significance levels are denoted as: ns (not significant, p >.05); *** (p <.001)

Table 3 Illustration of path weights with Estimation of R2

Effect	**Hypothesis**	**Path**	**Sign**	**β-value**	**Significance Level**	**R^2**	**Remarks**
Effect on ATT						**0.369**	
By PR	H1	PR → ATT	-	-0.143	ns		Not Supported
By PE	H2	PE → ATT	+	0.344	***(P <.001)		Supported
By EE	H3	EE → ATT	+	0.428	***(P <.001)		Supported
Effect on EE						**0.015**	
By FC	H4b	FC → EE	+	0.098	ns		Not Supported
Effect on BI						**0.530**	
By FC	H4a	FC → BI	+	-0.032	ns		Not Supported
By ATT	H5	ATT → BI	+	0.84	***(P <.001)		Supported
Effect on ACME						**0.365**	
By BI	H6	BI → ACME	+	0.575	***(P <.001)		Supported

The results confirm that user attitudes significantly affect their intentions and actual usage of ChatGPT in the medical educational context. However, FCs did not significantly influence EE or BI, suggesting that other factors not captured by the current model might play a role. This insight could be valuable for educational technology developers who focus on maximizing the usability and effectiveness of AI tools in medical education.

Discussion

ChatGPT's adoption in medical education is influenced by multiple factors that shape students' and educators' willingness to integrate AI-driven tools into their learning environments. While its potential benefits are widely acknowledged, adoption patterns vary based on contextual, methodological, and cultural factors, requiring a deeper analysis to understand the dynamics at play.

PE and EE are pivotal to the adoption of ChatGPT and other technologies in medical education. Our findings confirm this, as PE and EE were significantly associated with ATT toward ChatGPT, explaining 37% of the variance. This aligns with studies highlighting PE (or perceived usefulness) as a critical driver of adoption when users believe technology enhances learning outcomes and efficiency [26].

Among Moroccan nursing students, PE predicted ICT and social media use in education with EE as a moderating factor. Similarly, ChatGPT's ease of use likely fostered positive attitudes, reinforcing the importance of user-friendly interfaces in AI-assisted diagnostic tools [27]. However, the influences of PE and EE vary across educational settings. A study on mobile app adoption among medical students found that EE indirectly affected usage, whereas PE directly influenced behavioral intention [28].

Research on UTAUT in educational technology has identified reasons for the insignificant relationships between model constructs. For instance, 40% of respondents in a blockchain adoption study indicated that EE, facilitating conditions, and hedonic motivation did not influence adoption [29]. Similarly, a study on Google Classroom found that 35% of students reported that social influence, EE, and self-efficacy had no impact on their decision to use the platform [30]. Methodological limitations such as overlooking nonlinear relationships may contribute to these inconsistencies. A meta-analysis of 40 studies found that linear models explained only 60% of the technology adoption variance, compared to 75% when nonlinear relationships were considered [31]. These variations likely reflect contextual factors such as technological familiarity and cultural differences, underscoring the need for further research in medical education. In our study, the significant role of both PE and EE suggests that ChatGPT's perceived utility and ease of use align with Jordanian medical students' needs, thus supporting broader adoption.

PR, which is hypothesized to negatively affect attitudes, was not significant in our study. This is surprising, as PR is often a key barrier to technology adoption, particularly in healthcare education. Trust, security, and user familiarity shape risk perception with higher PR dampening adoption [32]. One study found that while 60% of students viewed AI tools as beneficial for assessments, 40% had concerns about data privacy and dependency [33]. Another study reported that 90% of students were familiar with AI tools, yet 35% lacked clarity on institutional AI policies, leading to uncertainty regarding proper use [34]. Additionally, overreliance on AI tools negatively affected problem-solving skills in 45% of students, while sparing use improved academic performance by 30% [35]. Addressing PR is crucial, as digital health literacy and user-centric designs can reduce apprehensions and enhance adoption [36].

FC was not significantly associated with BI or EE in our study, unlike other contexts where FCs strongly influence technology adoption. Research on mobile health applications, Moodle adoption, and mobile banking highlights the critical role of technical support, system compatibility, and resource availability in driving adoption [28, 37, 38]. However, in augmented reality (AR) adoption for primary education, FCs have an insignificant impact on BI, likely due to the novelty of technology and lack of supporting policies or training [39].

[28, 37]. The lack of significance in our study may reflect a cultural or contextual difference, where students and educators rely less on external support due to their familiarity with self-directed learning tools. However, ensuring accessible infrastructure and training remains essential for maximizing the potential of ChatGPT in medical education [40].

BI and its relationship with ACME demonstrated a clear pathway in our findings, which is consistent with UTAUT. ATT significantly influenced BI, which, in turn, predicted ACME, explaining 53% and 36.5% of the variance, respectively. These results align with broader UTAUT research, where ATT is a strong predictor of BI, particularly when technologies are perceived as useful and easy to use [41].

For ChatGPT, fostering positive attitudes by addressing usability concerns and demonstrating practical benefits can enhance its adoption. Cultural factors that shape technology acceptance in Middle East Hierarchical decision-making, communal learning, and gender norms may influence adoption behavior [42]. Barriers such as limited access to resources and institutional biases, particularly for women in STEM, highlight the need for equitable and culturally sensitive implementation strategies [43]. The COVID-19 pandemic further exacerbated digital literacy

gaps, emphasizing the need for targeted training and support to build trust in technologies, such as ChatGPT [44].

Comparative studies reveal regional differences in the adoption of the ChatGPT. Underdeveloped regions face infrastructural challenges [45], whereas developed nations focus on ethical frameworks and personalized learning [46]. In Jordan, addressing foundational barriers while fostering technological acceptance is the key to successful integration.

ChatGPT presents transformative opportunities in medical education but also raises concerns, particularly regarding misinformation. The risk of generating inaccurate or biased content poses challenges for critical assignments and research integrity [47]. These issues are compounded by a lack of training in ethical AI use, as seen in studies that highlight the risks of fabricated data and images [48].

Integrating AI literacy into curricula is essential for ensuring responsible use. Research emphasizes the need for critical evaluation skills to discern credible information and biases [49]. Coupled with the professional development of educators, these initiatives can promote ethical and effective ChatGPT use in medical education.

Our study underscores the potential of ChatGPT in medical education, if supported by tailored strategies. Addressing usability challenges, misinformation risks, and cultural apprehensions, while enhancing trust and digital literacy, can ensure successful integration. With institutional support, ChatGPT can revolutionize medical education and offer personalized learning, improved diagnostic training, and innovative teaching methods.

This study is the first to examine ChatGPT adoption in Jordanian medical education, thus filling a critical gap in Middle Eastern research. Using the UTAUT framework, it identifies the key factors influencing adoption among medical students and educators, reinforcing its relevance in understanding technology acceptance in medical education. Although our findings emphasize this potential, this study has several limitations. Reliance on self-reported data introduces a potential response bias, and the focus on a single institution limits generalizability. A larger sample size would enhance the statistical power. As the limited sample size and geographic scope restrict the generalizability of our findings, we recommend an international study across Middle Eastern countries to validate and extend these findings and better understand the regional adoption dynamics, cultural and infrastructural differences, and broader applications of ChatGPT in medical education. For the future, further tests could ensure the validity and reliability of our findings, a multi-collinearity test could identify and address potential collinearity issues among independent variables. Additionally, this study does not explore the physiological dimension related to the adoption of ChatGPT. Policymakers and institutions should consider developing training programs to improve the digital literacy among both the educators and the students, as well as establishing clear guidelines that address ethical concerns and the risk of misinformation, this could support the integration of ChatGPT into medical education. While this study focuses on Jordanian education, its findings and methodology have signification global implication, demonstrating that UTAUT is flexible model that can applied across different regions and cultures. Thus, this study serves as a key reference for AI adoption in medical education within cultural and global discussions. This pioneering study lays the groundwork for future studies on AI adoption in medical education, emphasizing the need for strategies to enhance usability and trust.

Conclusion

ChatGPT presents transformative opportunities for medical education in Jordan by enhancing personalized learning and streamlining research. Our findings highlight the pivotal role of attitudes influenced by performance and effort expectancies in driving adoption among medical students and educators. While perceived risk and facilitating conditions showed a minimal impact, trust-building measures and addressing misinformation were essential for sustained integration. By fostering positive attitudes and ensuring ethical use, ChatGPT can become a cornerstone of medical education by providing innovative, student-centered learning experiences.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12909-025-07336-y.

Supplementary Material 1

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Author contributions

Noura Alqaisi: Conceptualization, Writing– original draft. Sakhr Alshwayyat: Methodology, Writing– original draft. Saif Aburumman: Conceptualization, Writing– original draft. Nour Qassim: Formal analysis, Writing– original draft. Noor Almasri: Methodology, Writing– original draft. Fatima Algroosh: Formal analysis, Writing– original draft. Mesk Alkhatib: Methodology, Writing– original draft. Hamdah Hanifa: Writing– original draft, Writing– review & editing. Saif Aldeen AlRyalat: Supervision, Writing– review & editing. All authors have participated in writing the manuscript and reviewed the literature. All authors contributed to revision of the manuscript. All authors read and approved the final manuscript.

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Data availability

Data is provided within the manuscript or supplementary information files.

Declarations

Ethics approval and consent to participate

All procedures performed in this study involving human participants complied with the institutional and/or national research committee ethical standards and the 1964 Helsinki declaration and subsequent amendments or equivalent ethical standards. The study was designed and conducted in accordance with the ethical principles established by Jordan University Hospital. Therefore, ethical approval was obtained from the Institutional Review Board Committee, Jordan University Hospital. **ID number: 224000210.** Written informed consent was obtained from all the participants for the participation of this study and accompanying images. A copy of the written consent is available for review by the Editor-in-Chief of this journal on request.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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